

Python Programming Tasks & Instructions

- **TASK 1: Professional LinkedIn Announcement**
 - **Objective:** Share your official internship acceptance on your professional network.
 - **Requirements:** Post a high-resolution copy of your official **Internship Offer Letter** on LinkedIn. Craft a compelling caption detailing your learning goals, tag **Progree**, and use hashtags: `#pythonprogramming` `#scripting` `#software` `#progree`
- **TASK 2: Core Algorithmic Fibonacci Generation Module**
 - **Objective:** Build a Python module calculating exact sequence values under varying parameter limits.
 - **Requirements:** Write a clean Python function processing integer sequence bounds. Incorporate parameter sanitization handlers checking for negative input bounds, return outputs in structured list arrays, and output execution runtime benchmarks using the `timeit` framework.
- **TASK 3: Automated File Operating & Text-Parsing Script**
 - **Objective:** Build a background automation utility managing local directories and formatting text files.
 - **Requirements:** Leverage standard system libraries (`os`, `shutil`, `re`, or `pathlib`). Construct a script that parses an unorganized data folder, sorts files into subdirectories based on extensions, uses regular expressions to extract text strings (e.g., email patterns or transaction IDs) from flat logs, and logs outputs to a clean master CSV.
- **TASK 4: Mini Project - Production-Ready Multi-Intent Rule-Based Chatbot**
 - **Objective:** Program an advanced text-driven terminal conversational bot agent handling modular flows.
 - **Requirements:** Build a complex conversational application terminal loop. Apply basic natural language text normalization (case conversion, character stripping). Configure nested state logic handlers (`if-elif-else` structures combined with functional dictionary maps) to route user intent across multiple help trees, manage session state variations, and handle fallback error messages gracefully.